

The L2.5 Interlayer — UPL Reversible Influence Model (RIM)

A bounded zone between L2 (execution) and L3 (representation) where forward influence (chip $\rightarrow$ output) and reverse influence (output $\rightarrow$ chip) both produce named, measurable, causally traceable operators with explicit reversibility.

FORWARD F

$L_2 \rightarrow L_3$

Forward operators (F)

Representation:

cast_precision, quantize, normalize

Cache/Memory:

cache_pin, cache_evict, prefetch

Scheduler:

select_device, split_batch, fuse_kernel

Decoding:

sample_token, rerank, beam_prune

REVERSE B

$L_3 \rightarrow L_2$

Reverse operators (B)

Replay:

replay, rollback, restore_snapshot

Compensate:

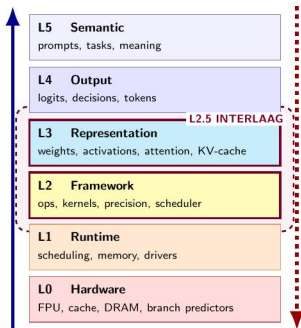
compensate, re-evaluate

Reroute:

reroute_compute, increase_precision

Stabilize:

stabilize_schedule, lock_determinism



L2.5 state $S_{L2.5}(t) = \langle C, H, M, O, R, P, N \rangle$

C compute (precision, kernel, batch, tensor layout) H hardware (CPU/GPU load, cache pressure, bandwidth)

M model (attention, embeddings, KV-cache, logits) O output (token, confidence, policy decision)

R reversibility class P provenance graph N UPL name

Forward mapping

$S_3 = F(S_2, Z)$

L_2 conditions and bridge events Z

determine L_3 representational state.

Reverse mapping

$S_2 = B(S_3, Z, \text{Policy})$

L_3 state and policy modify

subsequent L_2 execution.

Reversibility classes (per event)

EXACT full snapshot restores state

SNAPSHOT checkpoint-based restore

RECONSTRUCTABLE via provenance, bounded loss

COMPENSATING correction without restore

NON_REVERSIBLE no meaningful restore

Four reversal types (per claim)

state reversal exact state restoration

causal reversal traceability of cause

operational reversal rerun with new settings

semantic reversal output re-evaluation

UPL naming convention

UPL::<layer>::<direction>::<domain>::<operator>::<effect>

e.g. UPL::L2.5::FWD::PRECISION::FP32_TO_FP16::LOGIT_SHIFT

UPL::L2.5::REV::CONFIDENCE::LOW::GPU_REPLAY

Principle L2.5.Control: Every event between L2 and L3 is treated as a named, measurable, causally traceable, and possibly reversible operator. UPL adjudicates both forward and reverse claims under the same witness discipline.